

Exceptional Collection Note

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November 2025

Proposition 0.1. Let $X \hookrightarrow \mathbb{P}_{\mathbb{C}}^n$ be a smooth closed subscheme of dimension $m < n$. Then, $\mathcal{O}_X(-n), \dots, \mathcal{O}_X$ can never be an exceptional collection.

Proof. I want to exhibit non-vanishing for some

$$\mathrm{Ext}^i(\mathcal{O}_X, \mathcal{O}_X(-k)) = H^i(X, \mathcal{O}_X(-k))$$

for any i and $1 \leq k \leq n$. It seems to me that the least likely cohomology to vanish is $H^m(X, \mathcal{O}_X(-n))$, which by Serre duality is $H^0(X, \mathcal{O}_X(n) \otimes \omega_X)$. We argue that ω_X is n -regular, by showing that

$$H^i(X, \omega_X(n-i)) = 0,$$

for $i > 0$, which follows from Kodaira vanishing (since we only consider up to $i = m$, and $n > m$). So, $\omega_X(n)$ is globally generated, so being nonzero as a sheaf means that

$$H^0(X, \mathcal{O}_X(n) \otimes \omega_X) \neq 0.$$

Thus, the collection is not exceptional. □